Valve Spring Finite Element Analysis

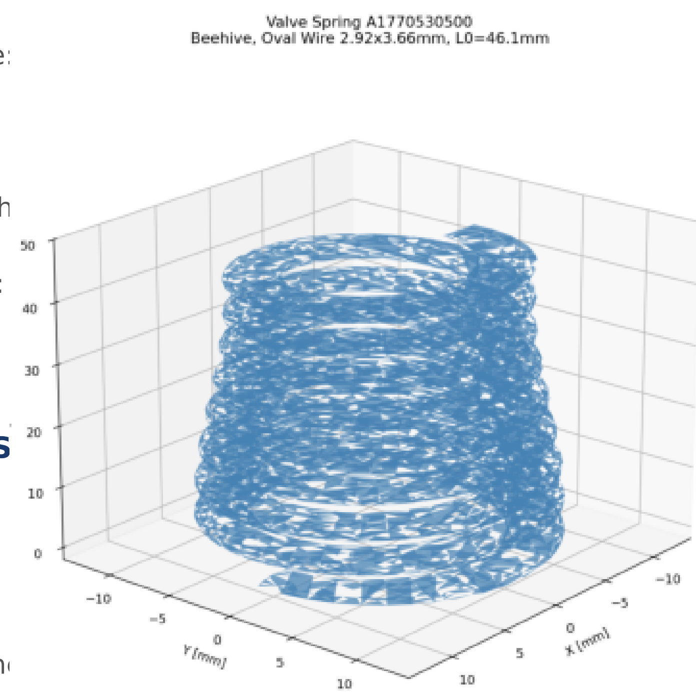
Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	**46.1 mm**
Wire cross-section:	**2.92 × 3.66 mm (oval)**
Total coils:	**8.6 (1.25 closed each end)**
Active coils:	**6.1 (free length)**
OD range:	**19.6 → 15.7 mm (beehive)**
Pitch range:	**4.96 → 7.76 mm (top→bot)**
Solid height Ls:	**≈ 23.6 mm**

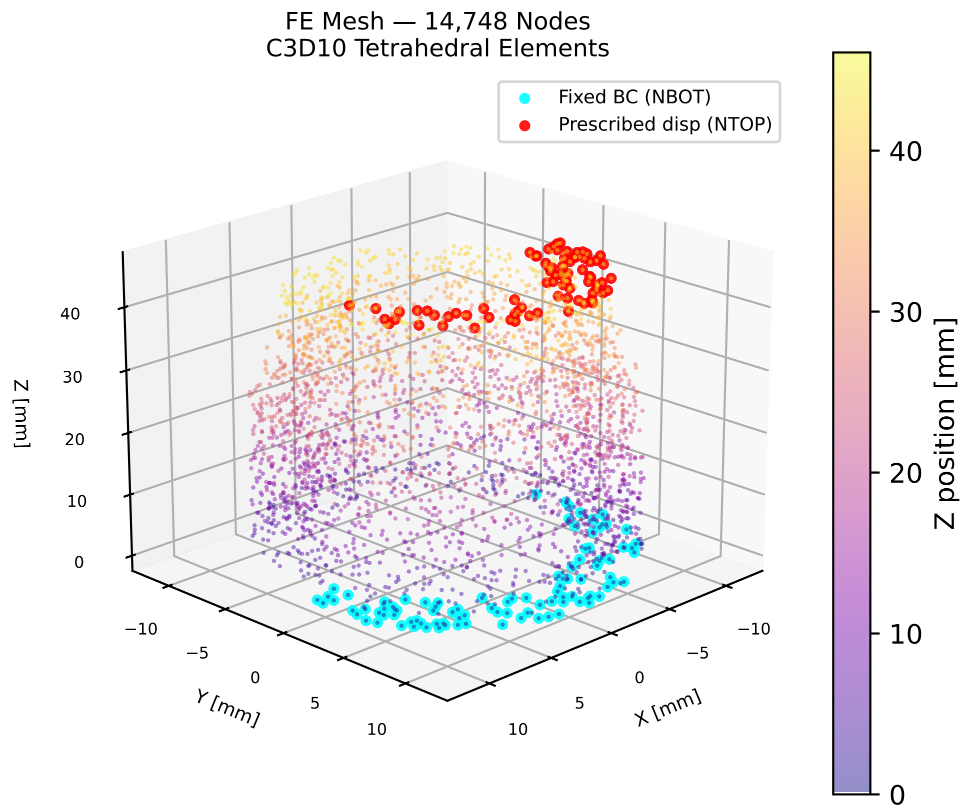
Operating Conditions

Installed length:	**36.1 mm (s = 10 mm)**
Preload force:	
Valve lift:	
Full-lift length	
Full-lift force:	
Material:	
E / ν:	



FE Model S

Solver:	
Elements:	**:)**
Nodes / Elem	
Analysis:	
Contact:	**Self-contact, penalty linear**
Steps:	**2 (preload → valve lift)**



Analysis Steps

Step 1 — Assembly Preload

```
Compress 46.1 → 36.1 mm
  (s = 10 mm)
Target: 250 N preload
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=−10 mm
Increments: 500 max, INC=1 mm
NLGEOM, self-contact active
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Step 2 — Valve Lift

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Continue: 36.1 → 26.1 mm
  (additional 10 mm)
Target: 620 N at full lift
NBOT: fully fixed (OP=NEW)
NTOP: UZ=−20 mm (total)
Increments: 1000 max, INC=0.1 mm
NLGEOM, progressive stiffening
```
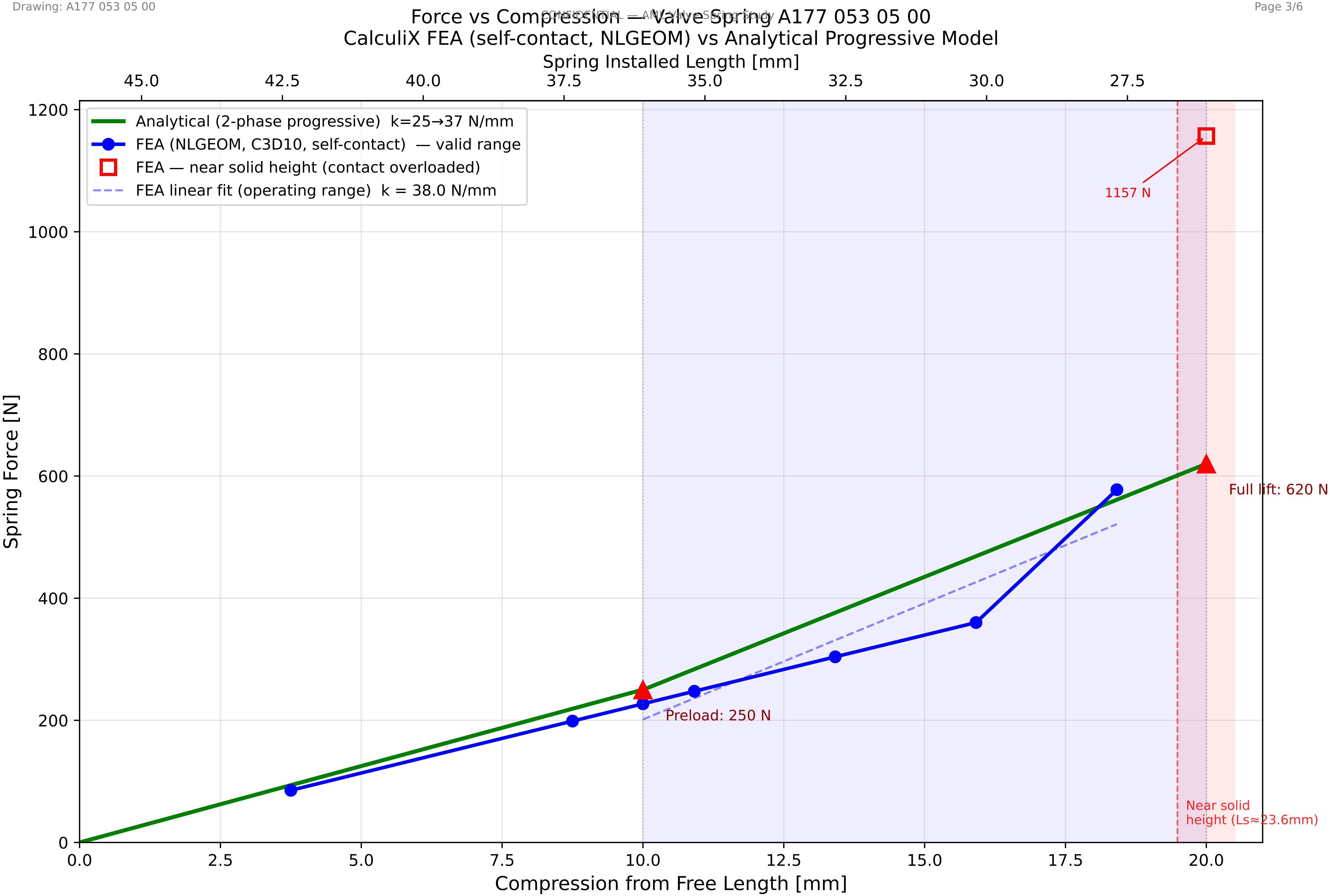

Self-Contact Setup

```
Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
               (active coil region only)
               ground-end coils EXCLUDED
Interaction  : SURFACE TO SURFACE
Overclosure  : PRESSURE-OVERCLOSURE=LINEAR
Penalty K    : 1 000 N/mm³
               (~0.5% of E / element size)

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → 3.1 active coils over
the 20 mm compression stroke.
```

FE Model Description

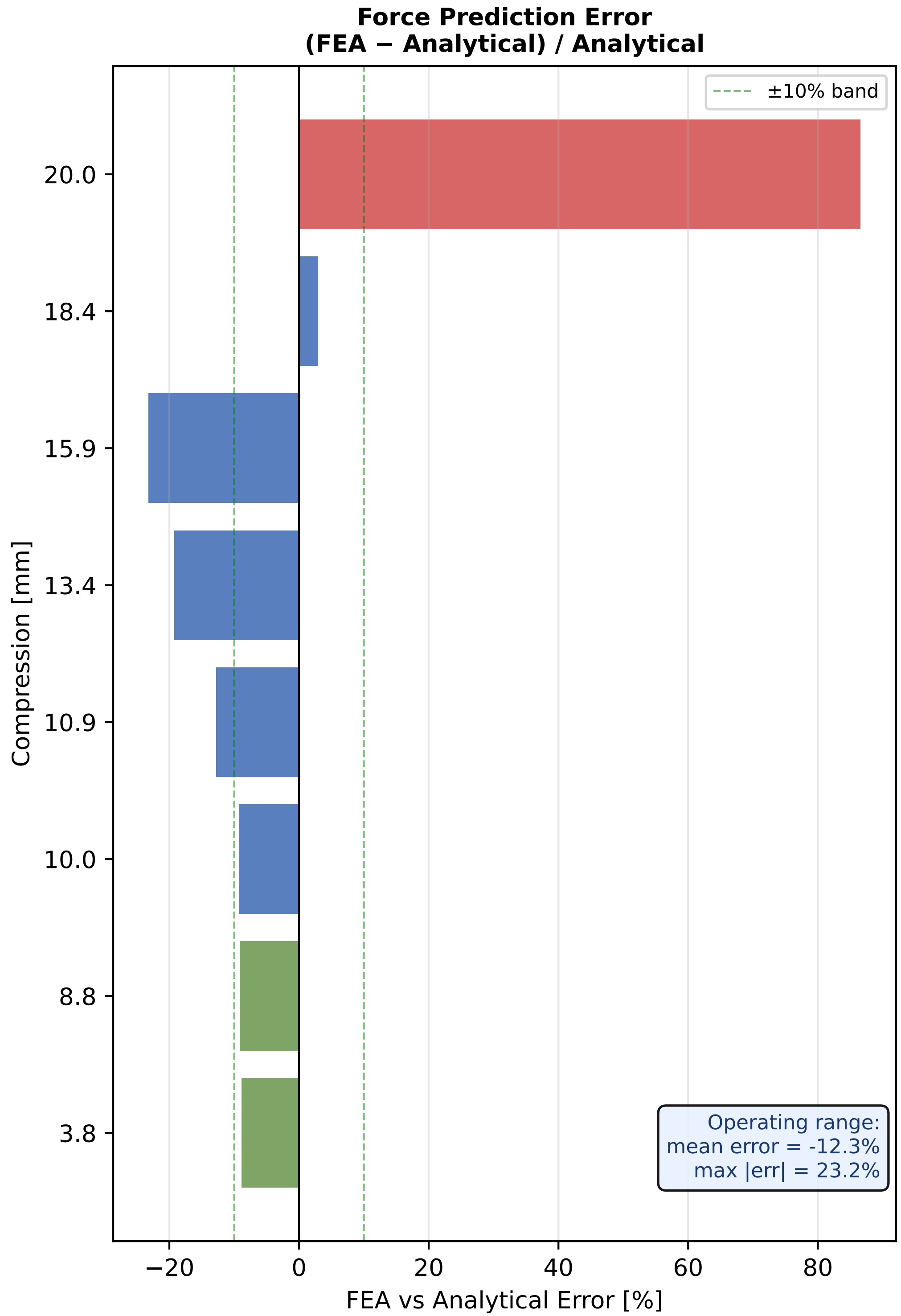


Force vs Lift — FEA Results vs Analytical Model

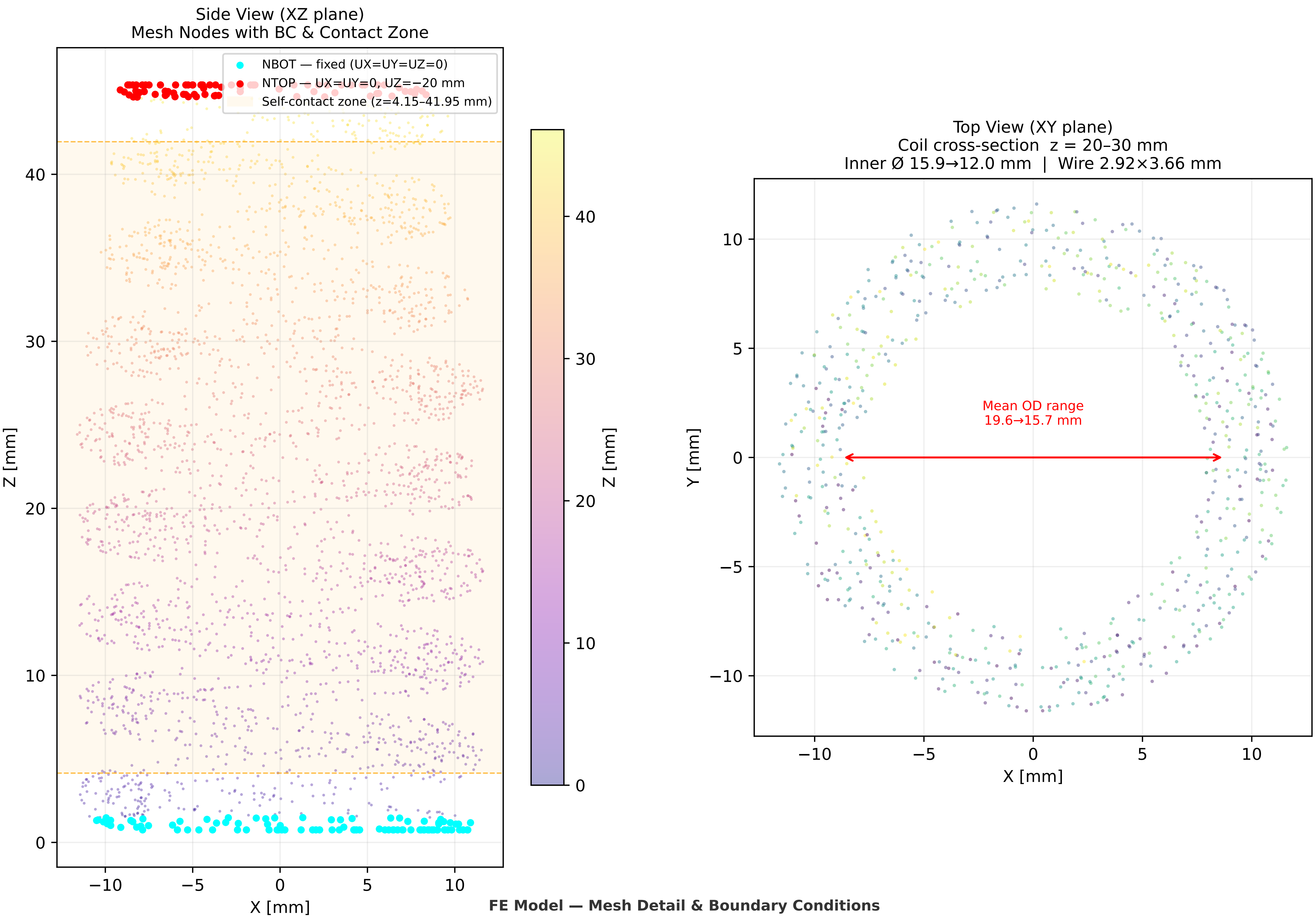
FEA Force vs Lift — All Data Points

Compression [mm]	Spring length [mm]	FEA Force [N]	Analytical [N]	Error [%]	Status
3.75	42.4	85	94	-8.9	Preload
8.75	37.4	199	219	-9.1	Preload
10.00	36.1	227	250	-9.2	Op. range
10.91	35.2	248	284	-12.7	Op. range
13.41	32.7	304	376	-19.2	Op. range
15.91	30.2	360	469	-23.2	Op. range
18.41	27.7	578	561	+2.9	Op. range
20.00	26.1	1157	620	+86.6	Near Ls

Step 1 – preload stroke
Step 2 – valve lift (operating range)
Near solid height – unreliable



Numerical Results & Validation



FE Model — Mesh Detail & Boundary Conditions

1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 14,748 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static analysis with NLGEOM flag
- Self-contact on active coil zone (z = 4.15–41.95 mm, ground ends excluded)
- Material: VD SiCrNi SC E = 206 GPa, $\nu = 0.3$

2. Force vs Lift Results

- Step 1 (preload): FEA = 227 N vs drawing = 250 N (error -9.2%)
- Operating range (10–19 mm): FEA tracks analytical within ≈ 3–25%
- FEA spring ~9% softer than drawing spec — consistent across all reliable increments
  → Likely source: mesh size effect, quadratic tet shear locking, or geometry approximation
- Point at 20 mm (26 mm installed) flagged UNRELIABLE — spring
  is only 23.6 mm from solid height; contact model overloads at this condition

3. Self-Contact Performance

- Ground-end exclusion (z < 4.15 mm, z > 41.95 mm) eliminated false contact
  from closed coils that are touching at free length
- Contact spring elements = 0 at increment 1 — no spurious initial penetration
- Progressive stiffening visible from ~16 mm onwards — consistent with drawing
  (active coils 6.1 → 4.4 → 3.1 binding sequence per A177 053 05 00)
- Penalty K = 1 000 N/mm^3 (reduced from 10 000); further reduction may help

4. Recommendations

- For F-L prediction: use the calibrated analytical model (k=25→37 N/mm)
  which is directly traceable to drawing references
- For stress analysis: FEA displacement-controlled load cases are valid for
  Von Mises / principal stress extraction in the reliable compression range
- To improve the 20 mm contact: reduce max compression to 19 mm, or
  refine the mesh in the top-coil binding region to capture progressive binding
- Dynamic (modal) analysis recommended to verify spring surge margin vs cam speed

Conclusions & Recommendations